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Toric geometry IV:

Functoriality & Smoothness

Last time Finished 'basic facts' about convex geometry.

Today Functoriality, Smoothness

Thm If (N, Σ) is a fan, then

(1) $X_\Sigma \rightarrow \operatorname{Spec}(\mathbb{Z})$ is flat, separated, and of finite presentation
Moreover, X_Σ is of relative dimension $\operatorname{rank}(N)$ over $\operatorname{Spec}(\mathbb{Z})$,
(Hence the same is true after every basechange.)

(2) For every ring A and cone $\sigma \in \Sigma$, the open embedding

$$X_{\sigma, A} \hookrightarrow X_{\Sigma, A}$$

is dense. In particular, there is a dense embedding of the torus $T_{N, A}$.

(3) There is a natural $T_{N, A}$ -action on $X_{\Sigma, A}$ making each embedding $X_{\sigma, A} \hookrightarrow X_{\Sigma, A}$ equivariant for the torus action.

Functoriality

Def. Let Fan denote the category with

(0) Objects: pairs (N, Σ) of a lattice N and a fan Σ in N .

(1) A morphism $(N, \Sigma) \longrightarrow (N', \Sigma')$ is a morphism of lattices $\phi: N \longrightarrow N'$ such that if $\sigma \in \Sigma$, then $\phi_{\mathbb{R}}(\sigma)$ is contained in a cone $\sigma' \in \Sigma'$.

Note. In the situation of (1), the assignment

$$\begin{aligned} \tilde{\phi}: \Sigma &\longrightarrow \Sigma' \\ \sigma &\longmapsto \min \{ \sigma' \in \Sigma' \mid \sigma' \supset \phi_{\mathbb{R}}(\sigma) \} \end{aligned}$$

is a map of posets.

Def. Let Sch^T be the category with

(0) Objects: triples $(T, X, a: T \times X \longrightarrow X)$ of an algebraic torus T , a scheme X , and a T -action $a: T \times X \longrightarrow X$.

(1) A morphism from (T, X, a) to (T', X', a') is a

pair of a morphism of group schemes $\phi: T \rightarrow T'$ and a morphism $f: X \rightarrow X'$ making the square

$$\begin{array}{ccc} T \times X & \xrightarrow{\alpha} & X \\ \phi \times f \downarrow & & \downarrow f \\ T' \times X' & \xrightarrow{\alpha'} & X' \end{array} \quad \text{commute}$$

Observation. The assignment $(N, \Sigma) \mapsto X_\Sigma$ defines a functor

$$\text{Fan} \longrightarrow \text{Sch}^{\mathbb{A}^1}$$

If $\phi: (N, \Sigma) \rightarrow (N', \Sigma')$ is a morphism of fans, the induced morphism

$$X_\Sigma = \text{colim}_{\sigma \in \Sigma} X_\sigma \longrightarrow \text{colim}_{\sigma' \in \Sigma'} X_{\sigma'} = X_{\Sigma'}$$

is induced by the map of posets $\tilde{\phi}: \Sigma \rightarrow \Sigma'$ and the universal property of the colimit.

Exercise. If (N, Σ) and (N', Σ') are fans, let $\Sigma \times \Sigma'$ be the fan in $N \times N'$ consisting of products $\sigma \times \sigma'$ for $\sigma \in \Sigma$ and $\sigma' \in \Sigma'$. Then

$$X_{\Sigma \times \Sigma'} \cong X_\Sigma \times X_{\Sigma'}$$

The functor of points of X_σ

Recollection. Let X be a scheme. The functor of points of X is the presheaf

$$\begin{aligned} X(-) : \text{Sch}^{\text{op}} &\longrightarrow \text{Set} && \text{'T-points of X'} \\ T &\longmapsto \text{Hom}_{\text{Sch}}(T, X) = X(T) \end{aligned}$$

represented by X . The functor of points is determined by its restriction to $\text{Aff}^{\text{op}} \simeq \text{CAlg}$, so we often refer to

$$\begin{aligned} X(-) : \text{CAlg} &\longrightarrow \text{Set} \\ A &\longmapsto X(A) := \text{Hom}_{\text{Sch}}(\text{Spec}(A), X) \end{aligned}$$

as the functor of points. If $X = \text{Spec}(B)$ is affine, then

$$X(A) = \text{Hom}_{\text{CAlg}}(B, A).$$

Ex. The functor of points of $\mathbb{P}_{\mathbb{Z}}^n$ is

$$\mathbb{P}_{\mathbb{Z}}^n(T) = \left\{ \begin{array}{l} \mathcal{L} \text{ line bundle on } T \\ + (s_0, \dots, s_n): \mathcal{O}_T^{\oplus (n+1)} \twoheadrightarrow \mathcal{L} \\ \text{surjection on stalks} \end{array} \right\} / \sim$$

Recall $\mathbb{Z} \langle - \rangle: \mathbf{CMon} \rightarrow \mathbf{CAlg}$ is left adjoint to the forgetful functor $A \mapsto (A, \cdot)$.

Ex. If S is a commutative monoid and $X = \text{Spec}(\mathbb{Z} \langle S \rangle)$, then for any ring A ,

$$\begin{aligned} X(A) &= \text{Hom}_{\mathbf{CAlg}}(\mathbb{Z} \langle S \rangle, A) \\ &= \text{Hom}_{\mathbf{CMon}}(S, (A, \cdot)) \end{aligned}$$

Ex. Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ a cone. Then X_{σ} has a natural $\mathbb{Z}$ -point $x_{\sigma}: \text{Spec}(\mathbb{Z}) \rightarrow X_{\sigma}$ corresponding to the homomorphism of commutative monoids

$$\mathbb{1}_{\sigma^{\perp}}: S_{\sigma} = \sigma^{\vee} \cap M \longrightarrow (\mathbb{Z}, \cdot)$$

given by

$$u \longmapsto \begin{cases} 1, & u \in \sigma^{\perp} \\ 0, & u \notin \sigma^{\perp} \end{cases}$$

To see that this is a monoid hom, note that $\sigma^{\perp}$ is a face of the cone $\sigma^{\vee}$. Hence if $u+v \in \sigma^{\perp}$, then $u, v \in \sigma^{\perp}$ (as explained last time). Thus the formula

$$\mathbb{1}_{\sigma^{\perp}}(u+v) = \mathbb{1}_{\sigma^{\perp}}(u) \mathbb{1}_{\sigma^{\perp}}(v)$$

holds.

Observe. The corresponding ring homomorphism $\mathbb{Z}\langle S_{\sigma} \rangle \rightarrow \mathbb{Z}$ is given by extending $\mathbb{1}_{\sigma^{\perp}}$ linearly. Since $1 \in \mathbb{Z}$ is in the image, this homomorphism is surjective. The kernel is

the ideal $\mathbb{Z}\langle S_\sigma \setminus \sigma^\perp \rangle \subset \mathbb{Z}\langle S_\sigma \rangle$. Since $\mathbb{Z}$ is a domain, this kernel is a prime ideal.

Exercise. Let $N = \mathbb{Z}^n$ and $\sigma = 0$. Identify x_0 as a point of the torus $T_N = (\mathbb{G}_m)^{\times n}$.

Exercise. Show that:

- (1) If σ spans $N_{\mathbb{R}}$, then x_σ is the unique $T(\mathbb{Z})$ -fixed point of $X_\sigma(\mathbb{Z})$.
- (2) If σ does not span $N_{\mathbb{R}}$, then there are no $T(\mathbb{Z})$ -fixed points of $X_\sigma(\mathbb{Z})$.

Smoothness

Recall. The property of a morphism of schemes $f: X \rightarrow Y$ being smooth is local on the source (as well as the target). There are many equivalent definitions. One is that $f: X \rightarrow Y$ is smooth if and only if:

(1) f is flat and of finite presentation

(2) For every geometric point $\bar{y} \rightarrow Y$, the geometric fiber $X_{\bar{y}}$ is regular.

Recall that a noetherian local ring $(A, \mathfrak{m})$ is regular if

$$\dim(A_{\mathfrak{m}}) = \dim_{k(\mathfrak{m})} T_{\mathfrak{m}}^{\vee} A,$$

i.e., the Zariski cotangent space $T_{\mathfrak{m}}^{\vee} A = \mathfrak{m}/\mathfrak{m}^2$ has the expected dimension. A scheme X is regular if X is locally noetherian and the local rings $\mathcal{O}_{X,x}$ at closed points are regular.

In particular A toric scheme X_Σ is smooth over $\text{Spec}(\mathbb{Z})$ if and only if all X_σ for $\sigma \in \Sigma$ are smooth over $\text{Spec}(\mathbb{Z})$.

Proposition Let N be a lattice of rank n and $\sigma \subset N_{\mathbb{R}}$ a cone of dimension d . The following are equivalent

- (1) The scheme X_σ is smooth over $\text{Spec}(\mathbb{Z})$ (hence the same is true after every basechange)
- (2) There are generators for σ (as a cone) that can be extended to a basis for the lattice N
- (3) There exists an isomorphism

$$X_\sigma \cong \mathbb{A}_{\mathbb{Z}}^d \times (\mathbb{G}_m)^{\times(n-d)}$$

Def. Let N be a lattice

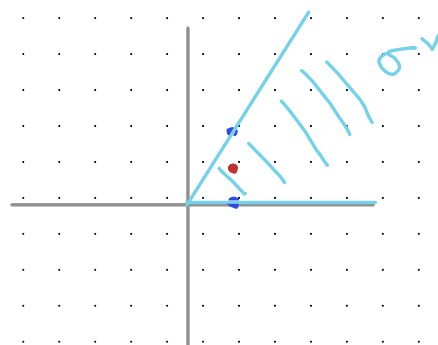
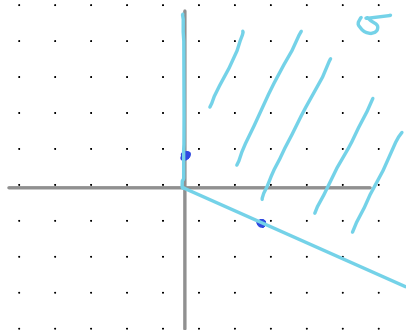
(1) A cone $\sigma \subset N_{\mathbb{R}}$ is smooth if there are generators for σ (as a cone) that can be extended to a basis for the lattice N .

(2) A fan Σ in N is smooth if every cone in Σ is smooth.

> Before proving the proposition, let's give an example demonstrating the necessity of (2).

> We also need some preliminaries we should have explained earlier.

Example Let $N = \mathbb{Z}^2$ and $\sigma = \text{cone}(e_2, 2e_1 - e_2)$.



Then $\sigma^\vee = \text{cone}(e_1^*, e_1^* + ze_2^*)$. We saw that X_σ is the quadratic cone

$$X_\sigma \cong \text{Spec}(\mathbb{Z}[x, y, z] / \langle y^2 - xz \rangle),$$

which has a singularity at the origin. The problem here is exactly that e_2 and $ze_1 - e_2$ don't generate $\mathbb{Z}^2$, they generate $\mathbb{Z} \times 2\mathbb{Z}$. This forces $\sigma^\vee \cap M$ to acquire an extra generator as well as relations.

Saturatedness & Splitting off a torus

Recollection. Let N be a lattice and $N' \subset N$ a sublattice. The following are equivalent:

- (1) The quotient N/N' is a lattice.
- (2) For $x \in N$, if there is a positive integer k such that $kx \in N'$, then $x \in N'$.

Def. If S is a commutative monoid, a submonoid $S' \subset S$ is saturated if (2) is satisfied.

Ex. If $\sigma \subset N_{\mathbb{R}}$ is a cone, then σ is saturated. Similarly, $S_{\sigma} = \sigma^{\vee} \cap M$ is saturated in M .

Observation. Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Let $N_{\sigma} = \sigma \cap N + (-\sigma \cap N)$

be the sublattice generated by σ . Then $\text{rank}(N_{\sigma}) = \dim(\sigma)$. Since σ is saturated, N_{σ} is saturated. Hence N/N_{σ} is a

lattice. Since N/N_σ is a projective abelian group, there exists a splitting

$$N \cong N_\sigma \oplus N''$$

Lemma. Let N be a lattice and $\sigma \subset N$ a cone. Choose a splitting $N \cong N_\sigma \oplus N''$, and write $\sigma = \sigma' \oplus 0$, where σ' is a cone in N_σ . Then this choice determines an isomorphism

$$X_\sigma \cong X_{\sigma'} \times T_{N''}$$

Proof.

Dually decompose M as $M' \oplus M''$ where $M' = (N_\sigma)^\vee$. Then

$$S_\sigma = ((\sigma')^\vee \cap M') \oplus M''$$

Since $\mathbb{Z} \langle - \rangle$ transforms direct sums into tensor products and Spec transforms tensor products into products, we conclude. □

Remark. More canonically, there is a fiber bundle

$$X_{\sigma'} \longrightarrow X_\sigma \longrightarrow T_{N/N_\sigma}$$

that splits, but does not have a canonical splitting.

A characterization of strong convexity

> We should have done this last lecture

Lemma Let V be a fd $\mathbb{R}$ -vector space and $\sigma \subset V$ a convex polyhedral cone. The following are equivalent

(1) σ is strongly convex

(2) $\sigma \cap (-\sigma) = 0$

(3) There is a $u \in \sigma^\vee$ with $\sigma \cap u^\perp = 0$, i.e., 0 is a face of σ .

(4) $\sigma^\vee$ spans $V^\vee$

Proof.

Note that $\sigma \cap (-\sigma)$ is the largest subspace of V contained in σ . So (1) and (2) are equivalent. The equivalence of (2) and (3) follows from the fact that $\sigma \cap (-\sigma)$ is the smallest face of σ . Finally, using the result from last time about properties of the isomorphism

$$\begin{array}{ccc} \text{Face}(\sigma)^{\text{op}} & \xrightarrow{\sim} & \text{Face}(\sigma^\vee) \\ \tau \mapsto & & \sigma^\vee \cap \tau^\perp, \end{array}$$

we see that

$$\dim(\sigma \cap (-\sigma)) + \dim(\sigma^\vee) = \dim(V)$$

So $\sigma \cap (-\sigma) = 0$ if and only if $\sigma^\vee$ spans V . □

Exercise Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ be a rational convex poly cone that spans $N_{\mathbb{R}}$. Show that $\sigma \cap N$ generates the lattice N as a group. That is,

$$N = \sigma \cap N + (-\sigma \cap N).$$

Note This does not say that there is a basis $v_1, \dots, v_n \in \sigma \cap N$ that also generates σ as a cone. Indeed, in the running example $N = \mathbb{Z}^2$ and $\sigma = \text{Cone}(e_2, 2e_1 - e_2)$ this isn't the case. The given integral cone generators are minimal.

However, $e_1 \in \sigma$, so it is true that $\sigma \cap \mathbb{Z}^2$ generates $\mathbb{Z}^2$ as a lattice.

Example Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a rational, strongly convex, polyhedral cone. Then $\sigma^\vee$ spans $M_{\mathbb{R}}$. So by the exercise, $S_\sigma = \sigma^\vee \cap M$ generates M as a group.

Proof of the characterization of Smoothness

Proof of Proposition

Clearly $(3) \Rightarrow (1)$. We'll prove that $(1) \Rightarrow (2) \Rightarrow (3)$ in two cases.

Case 1. σ spans $N_{\mathbb{R}}$.

In this case, $\sigma^\perp = 0$. We'll prove the claim by showing that if X_σ is smooth, then there are cone generators $v_1, \dots, v_n \in \sigma \cap N$ for σ that also form a basis for N , so

$$X_\sigma \simeq \mathbb{A}_{\mathbb{Z}}^n$$

Note that if X_σ is smooth, then for any separably closed field k , $X_{\sigma, k}$ is smooth. Note that x_σ basechanges to

define a closed point of $X_{\sigma, k}$ corresponding to the maximal ideal

$$m = \ker \left(k \langle S_\sigma \rangle \xrightarrow{k\text{-lin ext of } \mathbb{1}_{\sigma^\perp}} k \right)$$

Since $\sigma^1 = 0$, from what we explained earlier, we see that

$$m = k \langle S_\sigma - 0 \rangle.$$

Thus m^2 is the ideal in $k \langle S_\sigma \rangle$ generated

$$\begin{aligned} S_\sigma^{(2)} &= (S_\sigma - 0) + (S_\sigma - 0) \\ &= \left\{ \begin{array}{l} \text{sums of two nonzero} \\ \text{elements of } S_\sigma \end{array} \right\} \end{aligned}$$

Thus the cotangent space m/m^2 has basis images of

$$(S_\sigma - 0) \setminus S_\sigma^{(2)}, \text{ i.e., elements that are not sums of two vectors.}$$

> The smallest integral vectors along the edges of $\sigma^\vee$ are of this form.

Since $X_{\sigma, k}$ is smooth, in particular regular,

$$\dim_K(m/m^2) = \dim(X_\sigma) = n$$

In particular, $\sigma^\vee$ cannot have more than n edges, and the minimal generators in S_σ generate S_σ as a monoid. Since S_σ generates M as a group (explained earlier) and M has rank n , the minimal generators for S_σ form a basis for M as a lattice. By duality σ must then be generated by a basis for N as a lattice, as desired.

Case 2 $\dim(\sigma) < \text{rank}(N)$

Again we prove $(1) \Rightarrow (2) \Rightarrow (3)$. Let

$$N_\sigma := \sigma \cap N + (-\sigma \cap N)$$

be the sublattice of N generated by σ . As explained earlier, by choosing a splitting

$$N \simeq N_\sigma \oplus N''$$

and writing $\sigma = \sigma' \oplus 0$, we obtain an isomorphism

$$X_\sigma \cong X_{\sigma'} \times (\mathbb{G}_m)^{\times(n-d)}.$$

Since X_σ is smooth, so is $X_{\sigma'}$. Now σ' spans $(N_\sigma)_\mathbb{R}$, so by Case 1, we see that there are generators for σ' as a cone that are also a basis for N_σ as a lattice, and

$$X_{\sigma'} \cong \mathbb{A}_{\mathbb{Z}}^d.$$

Hence (2) and (3) hold, as desired. □